Dental X-rays and Risk of Meningioma

Context Ionizing radiation is a consistently identified and potentially modifiable risk factor for meningioma, the most frequently reported primary brain tumor in the United States. Objective To examine the association between dental x-rays, the most common artificial source of ionizing radiation, and risk of intra-cranial meningioma. Design and Setting Population-based case-control study design. Participants The study includes 1433 intra-cranial meningioma cases aged 29-79 years diagnosed among residents of the states of Connecticut, Massachusetts, North Carolina, the San Francisco Bay Area and eight Houston, Texas counties between May 1, 2006 and April 28, 2011 and 1350 controls that were frequency-matched on age, sex and geography. Main Outcome Measure The association of intra-cranial meningioma diagnosis with self-report of bitewing, full-mouth, and panorex dental x-rays. Results Over a lifetime, cases were more than twice (Odds ratio (OR) = 2.0, 95% confidence interval (CI), 1.4-2.9) as likely as controls to report having ever had a bitewing exam. Regardless of the age at which the films were received, persons who reported receiving bitewing films on a yearly or greater frequency had an elevated risk with odds ratios of 1.4 (95%CI: 1.0-1.8), 1.6 (95%CI: 1.2-2.0), 1.9 (95%CI: 1.4-2.6), and 1.5 (95%CI: 1.1-2.0) for ages <10, 10-19, 20-49, and 50+ years, respectively. Increased risk of meningioma was also associated with panorex films taken at a young age or on a yearly or greater frequency with persons reporting receiving such films under the age of 10 years at 4.9 times (95%CI: 1.8-13.2) increased risk of meningioma. No association was appreciated with location of tumor above or below the tentorium. Conclusion Exposure to some dental x-rays performed in the past, when radiation exposure was greater than in the current era, appears to be associated with increased risk of intra-cranial meningioma. As with all sources of artificial ionizing radiation, considered use of this modifiable risk factor may be of benefit to patients.